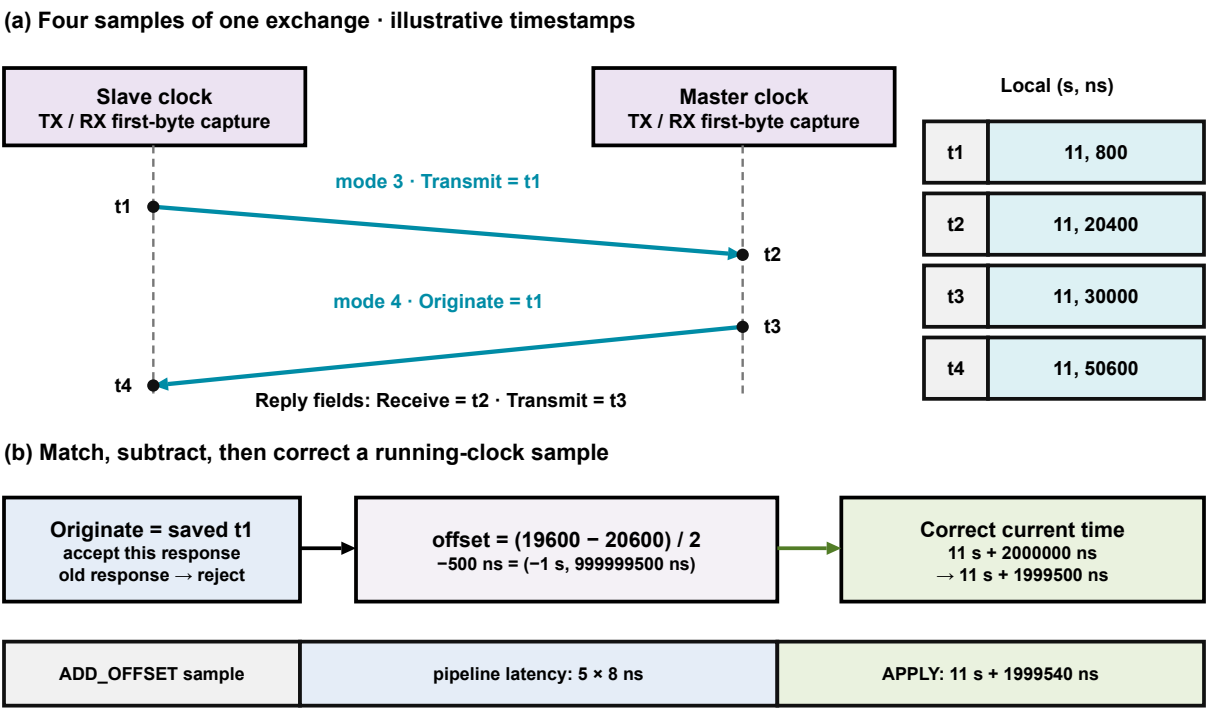
(a) Four samples of one exchange · illustrative timestamps
Slave clock
TX / RX first-byte capture
Master clock
TX / RX first-byte capture
mode 3 · Transmit = t1
t1
t2
mode 4 · Originate = t1
t3
t4
Reply fields: Receive = t2 · Transmit = t3
Local (s, ns)
t1
11, 800
t2
11, 20400
t3
11, 30000
t4
11, 50600
(b) Match, subtract, then correct a running-clock sample
Originate = saved t1
accept this response
old response → reject
offset = (19600 − 20600) / 2
−500 ns = (−1 s, 999999500 ns)
Correct current time
11 s + 2000000 ns
→ 11 s + 1999500 ns
ADD_OFFSET sample
pipeline latency: 5 × 8 ns
APPLY: 11 s + 1999540 ns